SiT: Scale-Interaction Transformer for Hyperspectral Image Classification

Zhonghao Chen, *Graduate Student Member, IEEE*, Swalpa Kumar Roy, *Senior Member, IEEE*, Hongmin Gao, *Senior Member, IEEE*, Yao Ding, and Bing Zhang, *Fellow, IEEE*

***Abstract*—Recently, transformers have emerged as a promising technique for capturing saliency feature dependencies that exist in hyperspectral (HS) images for the land-use and land-cover image classification tasks. However, existing transformer-based methods suffer from a significant challenge because they only take into account long-range dependencies between features in a fixed-size neighborhood. To tackle this problem, we proposed a novel scale-interaction transformer (SiT) tailored for HS image classification (HSIC) tasks. Specifically, two initial multiscale feature extraction modules are first designed for the modeling of spatial and spectral features, respectively. Furthermore, we investigate traditional transformer encoders to handle multiscale inputs and suggest a novel transformer-based encoder, called SiT encoder (SiTE), for modeling long-range dependencies of multiscale spectral and spatial features. More specifically, to enable the exploration of global correlation from multiscale perspectives in SiTE, an additional scale set is generated in addition to the query, key, and value sets. More importantly, to balance the contributions of spatial and spectral features during the final classification, we propose a novel loss function, called spectral and spatial consistency constraint (S^2C^2) loss, which adaptively fuses these features without introducing additional parameters. The experimental evaluations carried out on four public benchmarks showcase that the proposed SiT attains state-of-the-art (SOTA) classification performance and outperforms the previous SOTA by 2.83%, 3.04%, 2.67%, and 5.97% in terms of overall accuracy (OA) across four datasets, respectively. The code will be available at https://github.com/zhonghaochen/SiT**

***Index Terms*—Hyperspectral (HS) image classification (HSIC), multiscale feature, scale interaction, spectral and spatial consistency constraint (S^2C^2), transformer.**

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Zhonghao Chen is with the College of Information Science and Engineering, Hohai University, Changzhou 213200, China (e-mail: chenzhonghao@hhu.edu.cn).

Swalpa Kumar Roy is with the Department of Computer Science and Engineering, Alipurduar Government Engineering and Management College, Alipurduar, West Bengal 736206, India (e-mail: swalpa@agemc.ac.in).

Hongmin Gao is with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: gaohongmin@hhu.edu.cn).

Yao Ding is with the Laboratory of Intelligent Control, Rocket Force University of Engineering, Xi'an 710025, China (e-mail: dingyao.88@outlook.com).

Bing Zhang is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zb@radi.ac.cn).

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I. Introduction

HYPERSPECTRAL (HS) image, equipped with hundreds of spectral bands of land covers and their spatial distribution, holds great potential in the field of Earth observation (EO) [1], [2]. The rich spectral and spatial features not only enhance our understanding of the Earth's surface, but also offer valuable insights into diverse phenomena like environmental monitoring, precision agriculture, and urban planning [3], [4]. This underscores the attractiveness of HS image classification (HSIC) in contributing significantly to numerous EO applications [5], [6].

The initial research attention on HSIC was driven by the abundance of spectral features it offers, leading to the proposal of various related methods like support vector machines [7], random forests [8], and Markov random fields [9]. Despite their effectiveness, these methods face performance limitations due to the Hughes problem [10] arising from the high dimensionality of the spectra. To tackle this challenge, statistically driven approaches for spectral feature extraction and dimensionality reduction have been introduced and extensively adopted. Examples of such methods include principal component analysis (PCA) [11], linear discriminant analysis [12], and independent component analysis [13], which enhance the separability of samples. Further, researchers discovered that incorporating spatial features can mitigate the performance limitations resulting from the instability of spectral features caused by environmental factors. Among them, morphological profiles [14] and attribute profiles [15] have gained significant popularity as a tool for manually incorporating spatial features in HS images. Nevertheless, the limited discriminative power of manual features employed in these traditional methods does not sufficiently support higher accuracy classification.

In recent years, the capability of deep learning (DL) techniques in capturing complex and high-dimensional patterns within HS images is impressive, and numerous DL-based classification methods have been continuously explored [16]. Zhong et al. [17] introduced the deep belief network to accomplish HSIC with limited training samples. The autoencoder-based methods [18], [19] conduct self-supervised feature extraction of HS images, achieving promising outcomes. Moreover, generative adversarial network (GAN) [20]

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is further improved to accommodate HSIC tasks, with less risk of overfitting. Treating spectral features as sequential data, recurrent neural networks (RNNs) [21] are applied for the purpose of spectral feature extraction. Similarly, long short-term memory [22], alleviating the problem of gradient explosion that exists in RNNs, is adopted to HSIC. As an important branch of many DL architectures, convolutional neural networks (CNNs) have been widely studied for their excellent ability to capture neighborhood information and their applicability to HSIC tasks [23]. However, the limited receptive field size of the convolutional kernel restricts the interpretation of the global information of the land covers. Moreover, the transformer architecture [24] employs a multihead self-attention (MHSA) mechanism to allow the capture of dependencies of global tokens in a sequence. Encouraged by this, researchers tried to employ this backbone structure in vision-related tasks, and thus, the vision transformer (ViT) [25] was proposed. In the field of HSIC, transformer-based methods have gained attention by capturing long-range dependencies across spectral and spatial dimensions to learn discriminative representations [26]. However, there are still critical shortcomings in effectively learning robust global dependencies. The specific limitations of the aforementioned methods can be summarized as follows.

1) Simply using convolution kernels with varying receptive fields for multiscale spectral feature extraction often results in redundancy.
2) Most existing transformer-based methods in HSIC only consider long-range dependencies within fixed-size neighborhoods, lacking a full grasp of the global features essential for the task.
3) When adopting the approach of separately extracting spatial and spectral features and combining them through late fusion, achieving efficient fusion for final decision-making remains a critical yet often overlooked challenge.

Addressing the aforementioned concerns, a scale-interaction transformer (SiT) network is proposed to be applicable to HSIC. Briefly, the proposed SiT comprises three key parts. Initially, we execute the process of extracting multiscale spectral and spatial features separately, and a new multiscale spectral feature extraction strategy is proposed. Subsequently, to fully investigate the global correlation of obtained multiscale features, we developed an SiT encoder (called SiTE) based on a traditional transformer encoder by considering the comprehensive global correlations on multiscale perspectives. Moreover, rather than simply merging the extracted spectral and spatial features at the end of the network, which may lead to incorrect decisions due to error accumulation, we propose a fusion approach. This method naturally drives us to fuse spectral and spatial features by constraining the consistency of the fused features with the features to be fused, making the fused features produce more accurate results. Our major contributions are listed as follows.

1) A novel CNN-transformer-based multiscale feature extraction method, called SiT, is developed for HSIC by combining multiscale feature extraction with the transformer backbone.
2) We develop a multiscale spectral feature extraction module, i.e., direction-separated depthwise separable convolution block (D^2SCB), to specifically investigate multiscale spectral feature extraction by extracting adjacent and nonadjacent interband relationships rather than by simply changing the receptive field size.
3) We revisit conventional transformer encoders by considering how to receive multiscale features as the input and propose a new transformer-based encoder, SiTE, to learn long-range multiscale spectral and spatial features.
4) We introduce a spectral and spatial feature fusion scheme by designing a new loss function, i.e., spectral and spatial consistency constraint (S^2C^2) loss, to guide the fusion process and prevent undesirable fusion results.

II. Related Work

A. HSIC Methods Based on CNN

Benefiting from the weight-sharing and translation-invariance properties of convolutional operations, numerous CNN-based methods have gained widespread attention in the field of HSIC. Initially, Hu et al. [23] proposed a simple network by cascading 1-D convolution and max-pool operation to excavate diagnostic features only within the spectral domain. Makantasis et al. [27] further utilized neighborhood patches of pixel-to-be-classified as the input of the proposed 2-D CNN, with better performance due to the spectral–spatial features considered. After comparing 1-D CNN and 2-D CNN performance, Chen et al. [28] proposed the utilization of 3-D convolution operation to attain higher classification accuracy. To reduce the computational complexity, Ahmad et al. [29] proposed a shallow 3-D CNN to learn spectral–spatial features from dimension-reduced HS images. Moreover, as the expressive capability of the abstract feature yield by deeper networks becomes widely recognized, numerous deeper architectures have been explored for HSIC tasks. For example, Zhong et al. [30] proposed a deep 3-D CNN with residual connections to hierarchically extract spectral and spatial features. Li et al. [31] proposed a two-stream deep CNN to model local spectral–spatial and global spatial features simultaneously, and then, the fusion representation of the two deep features brings promising results. Meanwhile, to achieve a more comprehensive understanding of the land covers to be classified, the multiscale feature extraction strategy has been emphasized. In [32], the contextual deep CNN was proposed to model multiscale spectral–spatial features at the first layer. Yang et al. [33] proposed an enhanced multiscale feature fusion network to promote the interactions of features of different scales. Gao et al. [34] proposed the hierarchical shrinkage multiscale network by considering the variation of feature scale with network depth, achieving high accuracy classification results while halving the computational complexity. However, convolution operations cannot provide highly expressive global correlation embeddings for the pixels to be classified, whether from the spatial or spectral perspective. In recent years, HSIC

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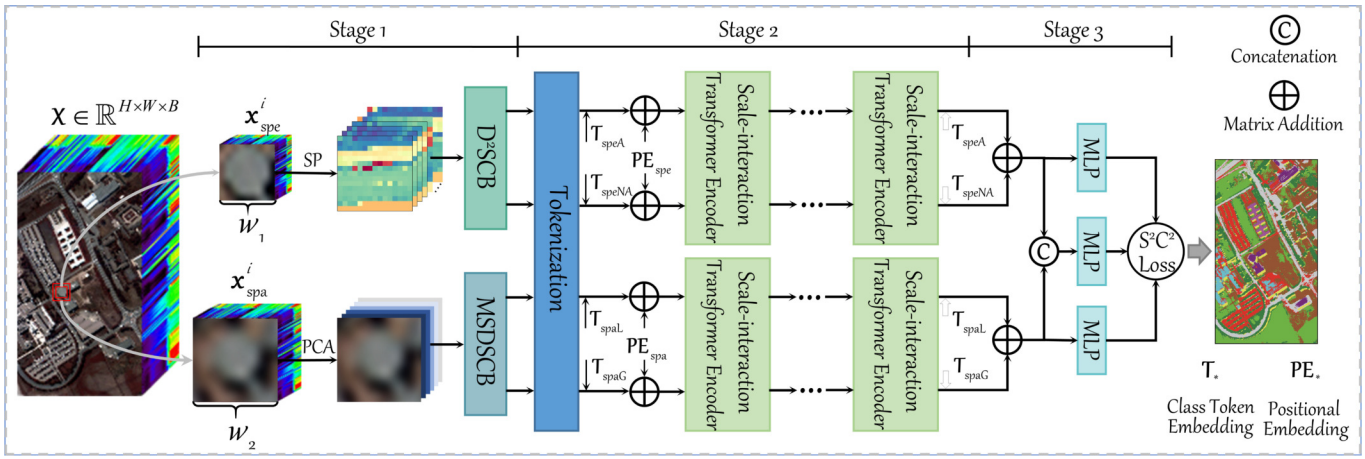


Fig. 1. Overview of the proposed SiT for the HSIC task, which consists of two branches for spectral and spatial feature extractions (the upper one is for spectral feature extraction and the lower one is for spatial feature extraction). The feature extraction process is designed in three stages. Stage 1: Spectral and spatial local multiscale features are extracted by the proposed D^2SCB and MSDSCB in parallel. Stage 2: Spectral and spatial global feature extraction in the multiscale perspective by the proposed SiTEs. Stage 3: Spectral and spatial feature fusions through the proposed S^2C^2 loss function.

methods have gradually introduced transformer encoders or their variants into the classification models.

B. HSIC Methods Based on Transformer

By delving deeper into the application of transformers in HSIC tasks, numerous efforts have been directed toward creating transformer-based methods to extract global feature representations from HS images [35]. In terms of spectral feature extraction, Hong et al. [36] investigated vanilla transformer encoders on global dependence extraction of spectral bands, providing a feasible application of transformer, for state-of-the-art (SOTA) HSIC. Inspired by [36], He et al. [37] proposed a semi-supervised classification method by adapting the transformer into the GAN architecture. For spectral–spatial feature extraction, ViT is being further explored, with numerous variants specifically proposed for the HSIC task. Sun et al. [38] employed the Gaussian-weighted strategy to tokenize shallow spectral–spatial features extracted by CNN as input of transformer encoders, designing the spectral–spatial feature tokenization transformer (SSFTT) for HSIC. He et al. [39] made BERT [40] applicable to HSIC tasks by taking advantage of BERT, which can model spatial features more flexibly. With similar incentives, Xue et al. [41] improved the transformer by introducing the spatial partition restore module to enhance emphasis on edge pixels. Furthermore, Song et al. [42] proposed a bottleneck spectral–spatial transformer (BS2T), which adopts the CNN module to model spectral and spatial shallow features separately and then sends them into the transformer encoders. In [43], dual types of context were hierarchically considered by the transformer encoder, achieving commendable intraclass local and global consistency predictions. More robustly, Roy et al. [44] employed spectral and spatial morphological convolutions to generate more discriminative tokens as the input of the transformer encoders. Similarly, Peng et al. [45] considered global spatial and spectral features, respectively, and correspondingly proposed a cross-attention module to investigate the mutual complementarity between them. Moreover, Wang et al. [46] enhanced a transformer-based model by designing a novel MHSA block, achieving a robust HSIC framework while reducing computational complexity. However, the existing transformer-based approaches often fail to explore the correlations between features at different scales from a global perspective.

In this work, our proposed SiT aims to generate highly representative multiscale spatial and spectral local features and explore their global correlation using the customized SiTE. Moreover, we adaptively fuse spatial and spectral features to harness their complementarity during prediction, achieving superior classification performance.

III. Methodology

A. Overview of the Proposed SiT

As illustrated in Fig. 1, the proposed HS image classifier is developed in a dual path with three-stage manner to better investigate the spectral–spatial feature representation. Routinely, given an HS image $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where the $H \times W$ represents the spatial size and B represents the number of spectral channels. We first crop the original HS image into two different sizes (i.e., w_1 and w_2, $w_1 < w_2$) of cubes centered on each pixel, i.e., $\mathbf{x}^i_{\text{spe}} \in \mathbb{R}^{w_1\times w_1\times B}$ and $\mathbf{x}^i_{\text{spa}} \in \mathbb{R}^{w_2\times w_2\times B}$, where $i \in (1, 2, \ldots, HW)$. For the spectral extraction path, the spectral information from the smaller local neighborhood of the pixel-to-be-classified (i.e., $\mathbf{x}^i_{\text{spe}}$) is considered. Similarly, the spatial features are excavated from a bigger local neighborhood.

Moreover, the process of spectral–spatial feature extraction can be described in three stages. In the first stage, the local multiscale spectral or spatial feature embeddings are extracted separately. In the second stage, the proposed SiTE not only models the long-range correlation of features at different scales individually, but also explores the global correlation from a multiscale perspective. After undergoing the second phase, the high discriminative spectral and spatial features are generated. Empirically, in the last stage, we add the above-obtained spectral and spatial features together to get the fused spectral–spatial feature. Notably, an S^2C^2 loss function is developed to coordinate consistency between the spectral and

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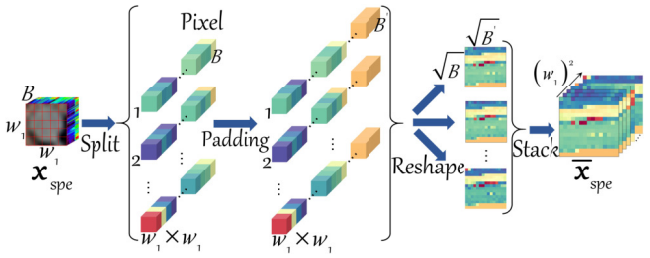


Fig. 2. Diagram of the SP operation in the proposed SiT. We first split $\mathbf{x}_{spe}$ into $w_1 \times w_1$ pixel spectral vectors with the shape of $1 \times B$ along the spatial dimension. Then, to meet the dimensionality requirement of enabling the reshaping of each spectral vector into a square matrix, we pad p zero values along the spectral dimension of each spectral vector. After that, we can reshape the padded spectral vector into the spectral matrix with a size of $(B')^{1/2} \times (B')^{1/2} \times 1$, where $B' = B + p$. Finally, we stack all spectral matrices reshaped from each spectral vector along the channel dimension to generate a new spectral cube.

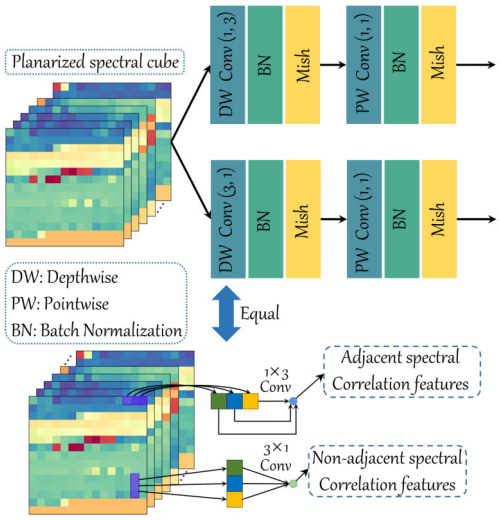


Fig. 3. Diagram of the D^{2}SCB used on the top-planarized spectral cube.

spatial features in the predicted results and the fused features, ensuring the representational capacity of the fused features.

B. Stage 1: Multiscale Feature Extraction

1) Spectral Multiscale Feature Extraction: The hundreds of spectral bands bring abundant information but also pose a challenge for the extraction of highly discriminative features [47]. In a nutshell, richer but complex correlated information always exists within high-dimensional data. For HS images, interband relationships across different spectral ranges can often yield a more comprehensive feature representation, which is a crucial aspect for accurately identifying the land-cover materials. To do this, inspired by our previous work [48], we first introduce the spectral planarization (SP) strategy to facilitate the exploration of spectral features at different scales. Then, the D^{2}SCB is proposed to extract multiscale spectral features for HSIC tasks.

Usually, the multiscale spectral interband correlation embedding $\boldsymbol{\varepsilon}$ is obtained by applying convolution operation along spectral dimension with varying sizes of kernels for the given input $\mathbf{x}_{\text{spe}}$, which can be formulated by the following equation:

$$\varepsilon = \sum \text{Conv_}k\left(\mathbf{x}_{\text{spe}}\right) \tag{1}$$

where Conv_k$(\cdot)$ denotes the convolutional operation with varying kernel of size $(1, 1, k)$ while k represents the size of receptive filed. Obviously, this type of strategy inevitably leads to certain ranges of cross-band features being extracted repeatedly, which brings about redundancy between the extracted spectral features at different scales. Therefore, we propose a simple and generic strategy to obtain multiscale spectral features more effectively, which we have abbreviated as SP and D^{2}SCB. First of all, the SP operation is performed on $\mathbf{x}_{\text{spe}}$. As illustrated in Fig. 2, we first split $\mathbf{x}_{\text{spe}}$ into $w_1 \times w_1$ pixel spectral vectors with the shape of $1 \times B$ along the spatial dimension. Then, in order to meet the dimensionality requirement so that each spectral vector can be reshaped into a square matrix, we also use p zero values padding along the spectral dimension of each spectral vector. Since the number of spectral bands varies across different datasets, the size of padding (i.e., p) also varies accordingly, which can be calculated as follows:

$$p = \left(\lceil \sqrt{B} \rceil\right)^2 - B \tag{2}$$

where $\lceil \cdot \rceil$ represents the ceiling function. After that, we can reshape the padded spectral vector into the spectral matrix with a size of $(B')^{1/2} \times (B')^{1/2} \times 1$, where $B' = B + p$. Finally, we stack all spectral matrices that have been reshaped from each spectral vector along the channel dimension to generate a new spectral cube, i.e., $\bar{\mathbf{x}}_{\text{spe}} \in \mathbb{R}^{(B')^{1/2} \times (B')^{1/2} \times (w_1)^2}$, for further multiscale spectral feature extraction. Benefiting from the aforementioned SP operation, the planarized spectral cubes allow multiscale features to be extracted flexibly. Specifically, our proposed D^{2}SCB extracts features between spectral bands at different scales by extracting adjacent and nonadjacent band features in the horizontal and vertical directions of the planarized spectra, respectively. To this end, high discriminative and low redundant multiscale spectral features can be generated. As shown in Fig. 3, the D^{2}SCB is composed of two 2-D depthwise separable (DS) convolutional operations with kernels of sizes 1×3 and 3×1, respectively, which are performed in parallel but in different directions. After each DS convolutional operation, a batch normalization (BN) layer followed by a Mish activation [49] layer is applied to facilitate model fitting. Interestingly, the adopted Mish activation improves model stability more than the rectified linear unit (ReLU) function, which can also be mathematically formulated as $\text{Mish}(\mathbf{x}) = \mathbf{x} \times \tanh(\ln(1 + e^{\mathbf{x}}))$, where $\mathbf{x}$ is the input of the activation function. As delineated in Fig. 4, the spectral signatures associated with local adjacent spectral bands denoted as $\mathbf{M}_{\text{speA}} \in \mathbb{R}^{(B')^{1/2} \times (B')^{1/2} \times (w_1)^2}$ can be well-captured by the upper branch of D^{2}SCB while the lower branch can produce nonadjacent spectral interband features in parallel and represented as $\mathbf{M}_{\text{speNA}} \in \mathbb{R}^{(B')^{1/2} \times (B')^{1/2} \times (w_1)^2}$. As a result, the multiscale spectral feature can be fully represented when both the SP method and D^{2}SCB are employed hierarchically.

2) Spatial Multiscale Feature Extraction: As shown in Fig. 1, for the given input $\mathbf{x}_{\text{spa}}$, the PCA is adopted to preserve

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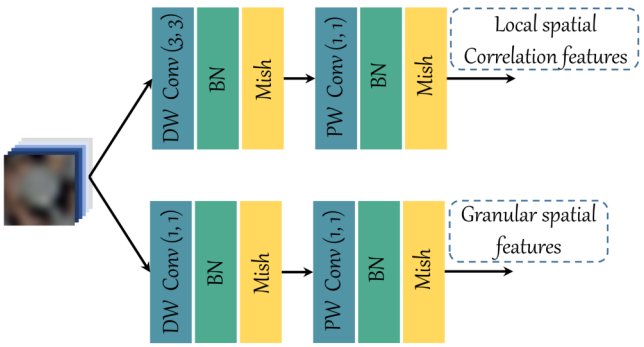


Fig. 4. Diagram of the MSDSCB for the spatial feature extraction.

P principle components and reduce the dimension of the $\mathbf{x}_{\text{spa}}$ from B to P, yielding $\bar{\mathbf{x}}_{\text{spa}} \in \mathbb{R}^{w_2 \times w_2 \times P}$. Then, in order to explore the spatial multiscale features, similar to the above spectral feature extraction module, we also adopt DS convolution to construct the multiscale DS convolutional block (MSDSCB) (see Fig. 4). Specifically, two 2-D DS convolutions with kernel of sizes 3×3 and 1×1, respectively, are applied on $\bar{\mathbf{x}}_{\text{spa}}$ to model the local and spatially correlated feature and granular information separately, denoted as $\mathbf{M}_{\text{spaL}} \in \mathbb{R}^{w_2 \times w_2 \times P}$ and $\mathbf{M}_{\text{spaG}} \in \mathbb{R}^{w_2 \times w_2 \times P}$. It is also observed that the BN layer and Mish activation layer are applied after the convolution operation in the proposed MSDSCB.

C. Stage 2: Global Attention and Scale Interaction

Four distinct scale embedded features from spectral and spatial profiles, i.e., $\mathbf{M}_{\text{speA}}$, $\mathbf{M}_{\text{speNA}}$, $\mathbf{M}_{\text{spaL}}$, and $\mathbf{M}_{\text{spaG}}$, are obtained after the successful implementation of D^{2}SCB and MSDSCB, respectively. However, these features are still limited in reflecting global correlations. Thus, we can further utilize the transformer encoder structure to establish the interdependencies between these obtained local features. Given the shape of these four feature maps, a ViT-based model is naturally adopted to account for extracting their global features. However, a straightforward use would ignore global correlations in a multiscale perspective, as the encoder of vanilla ViT can only handle a single input at a time. Therefore, to explore such meaningful correlations during the learning of long-range dependencies, a novel SiTE is tailored for multiscale long-range spectral and spatial feature extractions.

1) Spectral and Spatial Feature Tokenizations: According to the working principle of ViT, to satisfy the requirement, the transformer accepts the sequence of patches as input, and in the initial stage, the process involves preprocessing the previously acquired feature maps accordingly. For the given $\mathbf{M}_{\text{speA}}$ and $\mathbf{M}_{\text{speNA}}$, we rearrange them into two sequences of flattened 2-D patches, represented by $\mathbf{M}'_{\text{speA}} \in \mathbb{R}^{N_{\text{spe}} \times (s^2 \times w_1^2)}$ and $\mathbf{M}'_{\text{speNA}} \in \mathbb{R}^{N_{\text{spe}} \times (s^2 \times w_1^2)}$, where s is set to 1. Thus, the length of spectral sequence is $N_{\text{spe}} = (B')^{1/2} \times (B')^{1/2}$ and the dimension of each patch within $\mathbf{M}'_{\text{speA}}$ or $\mathbf{M}'_{\text{speNA}}$ is $w_1 \times w_1$. Next, a trainable linear projection layer is used to map the resulting patches into a D-dimensional space in order to satisfy the transformer's requirement so that the latent features be of dimension D, which can be formulated as follows:

$$\tilde{\mathbf{M}}_{\text{speA}} = \mathbf{M}'_{\text{speA}} \times \boldsymbol{\omega} \tag{3a}$$

$$\tilde{\mathbf{M}}_{\text{speNA}} = \mathbf{M}'_{\text{speNA}} \times \boldsymbol{\omega} \tag{3b}$$

where $\boldsymbol{\omega} \in \mathbb{R}^{(w_1)^2 \times D}$ represents the linear transformation and $\tilde{\mathbf{M}}_{\text{speA}} \in \mathbb{R}^{N_{\text{spe}} \times D}$ and $\tilde{\mathbf{M}}_{\text{speNA}} \in \mathbb{R}^{N_{\text{spe}} \times D}$ are the two spectral tokens embedded features of different scales. The feature tokenization process is performed according to (3a) and (3b); then, we empirically concatenate the class tokens $\mathbf{T}_{\text{speA/NA}} \in \mathbb{R}^{1 \times D}$ and next added the position embedding $\mathbf{PE}_{\text{spe}} \in \mathbb{R}^{(N_{spe}+1) \times D}$ to spectral tokens individually before fed to the following transformer encoders, that is:

$$\mathbf{z}_{\text{speA}} = \left[\mathbf{T}_{\text{speA}}; \tilde{\mathbf{M}}^1_{\text{speA}}; \tilde{\mathbf{M}}^2_{\text{speA}}; \ldots; \tilde{\mathbf{M}}^{N_{\text{spe}}}_{\text{speA}}\right] + \mathbf{PE}_{\text{spe}} \tag{4a}$$

$$\mathbf{z}_{\text{speNA}} = \left[\mathbf{T}_{\text{speNA}}; \tilde{\mathbf{M}}^1_{\text{speNA}}; \tilde{\mathbf{M}}^2_{\text{speNA}}; \ldots; \tilde{\mathbf{M}}^{N_{\text{spe}}}_{\text{speNA}}\right] + \mathbf{PE}_{\text{spe}} \tag{4b}$$

where $\mathbf{z}_{\text{speA}} \in \mathbb{R}^{(N_{\text{spe}}+1) \times D}$ and $\mathbf{z}_{\text{speNA}} \in \mathbb{R}^{(N_{\text{spe}}+1) \times D}$ represent the input sequence of tokens obtained for the SiTEs.

In a similar fashion, for the given $\mathbf{M}_{\text{spaL}}$ and $\mathbf{M}_{\text{spaG}}$, we can obtain the multiscale spatial feature tokens, i.e., $\tilde{\mathbf{M}}_{\text{spaL}} \in \mathbb{R}^{N_{\text{spa}} \times D}$ and $\tilde{\mathbf{M}}_{\text{spaG}} \in \mathbb{R}^{N_{\text{spa}} \times D}$, by employing a similar tokenization process as discussed in (3a) and (3b), respectively. Once the spatial feature tokenization process is completed, the spatial class tokens $\mathbf{T}_{\text{spaL/G}} \in \mathbb{R}^{1 \times D}$ are concatenated with the spatial tokens, followed by the addition of position embedding $\mathbf{PE}_{\text{spa}} \in \mathbb{R}^{(N_{\text{spa}}+1) \times D}$, which can be specifically formulated as follows:

$$\mathbf{z}_{\text{spaL}} = \left[\mathbf{T}_{\text{spaL}}; \tilde{\mathbf{M}}^1_{\text{spaL}}; \tilde{\mathbf{M}}^2_{\text{spaL}}; \ldots; \tilde{\mathbf{M}}^{N_{\text{spa}}}_{\text{spaL}}\right] + \mathbf{PE}_{\text{spa}} \tag{5a}$$

$$\mathbf{z}_{\text{spaG}} = \left[\mathbf{T}_{\text{spaG}}; \tilde{\mathbf{M}}^1_{\text{spaG}}; \tilde{\mathbf{M}}^2_{\text{spaG}}; \ldots; \tilde{\mathbf{M}}^{N_{\text{spa}}}_{\text{spaG}}\right] + \mathbf{PE}_{\text{spa}} \tag{5b}$$

where $\mathbf{z}_{\text{spaL}} \in \mathbb{R}^{(N_{\text{spa}}+1) \times D}$, $\mathbf{z}_{\text{spaG}} \in \mathbb{R}^{(N_{\text{spa}}+1) \times D}$ represents the obtained output sequence of tokens for the subsequent SiTEs, and $N_{\text{spa}} = w_2 \times w_2$.

2) Scale-Interaction Multihead Attention: To comprehensively investigate the global correlation of tokens from a multiscale perspective, as shown in Fig. 5, we propose the SiTE equipped with a newly developed scale-interaction multihead attention (SIMHA) module. More specifically, we introduce a multiscale global feature weighting operation between two parallel self-attention processes within the proposed SIMHA, which adaptively learns the global correlations of tokens with multiscale information, in addition to extracting global correlations of tokens from the perspective of single scale. It can also be noted that, in addition to generating query, key, and value sets, we also produce a scale set ($\mathbf{S}$) specifically for producing comprehensive multiscale representations. Taking the spectral multiscale sequence tokens $\mathbf{z}_{\text{speA}}$, and $\mathbf{z}_{\text{speNA}}$ as an example, and the process of SIMHA will be presented hierarchically and its inputs (denoted as $\mathbf{Q}_{\text{A/NA}}$, $\mathbf{K}_{\text{A/NA}}$, $\mathbf{V}_{\text{A/NA}}$, and $\mathbf{S}_{\text{A/NA}}$) are initially formulated as

$$\begin{aligned} \mathbf{Q}_{\text{A}} &= \mathbf{z}_{\text{speA}}\boldsymbol{\omega}_q, \mathbf{K}_{\text{A}} = \mathbf{z}_{\text{speA}}\boldsymbol{\omega}_{\text{k}}, \\ \mathbf{V}_{\text{A}} &= \mathbf{z}_{\text{speA}}\boldsymbol{\omega}_{\text{v}}, \mathbf{S}_{\text{A}} = \mathbf{z}_{\text{speA}}\boldsymbol{\omega}_{\text{s}} \end{aligned} \tag{6}$$

$$\begin{aligned} \mathbf{Q}_{\text{NA}} &= \mathbf{z}_{\text{speNA}}\boldsymbol{\omega}_{\text{q}}, \mathbf{K}_{\text{NA}} = \mathbf{z}_{\text{speNA}}\boldsymbol{\omega}_{\text{k}} \\ \mathbf{V}_{\text{NA}} &= \mathbf{z}_{\text{speNA}}\boldsymbol{\omega}_{\text{v}}, \mathbf{S}_{\text{NA}} = \mathbf{z}_{\text{speNA}}\boldsymbol{\omega}_{\text{s}} \end{aligned} \tag{7}$$

where $\boldsymbol{\omega}_{\text{q}} \in \mathbb{R}^{D \times d}$, $\boldsymbol{\omega}_{\text{k}} \in \mathbb{R}^{D \times d}$, $\boldsymbol{\omega}_{\text{v}} \in \mathbb{R}^{D \times d}$, and $\boldsymbol{\omega}_{\text{s}} \in \mathbb{R}^{D \times d}$ denote the linear projection weights and d represents the dimension of the output feature vector of the linear projection

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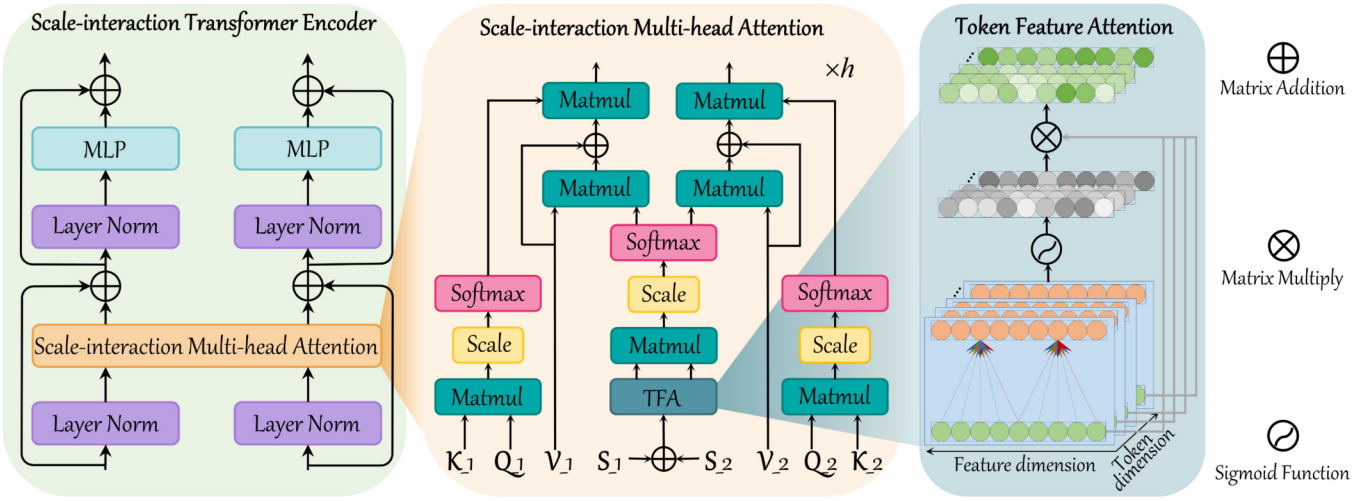


Fig. 5. Diagram of the proposed SiTE, which is equipped with a newly developed SIMHA. It can receive two feature maps with different scale information and hierarchically learn the global correlation of tokens from both single and multiscale perspectives. The newly proposed TFA is dedicated to the weighted fusion of two token sets (i.e., $\mathbf{S}_1$ and $\mathbf{S}_2$) with different scale features along the feature dimension.

layer. Subsequently, the query ($\mathbf{Q}_{\mathrm{A/NA}} \in \mathbb{R}^{(N_{\mathrm{spe}}+1)\times d}$), key ($\mathbf{K}_{\mathrm{A/NA}} \in \mathbb{R}^{(N_{\mathrm{spe}}+1)\times d}$), value ($\mathbf{V}_{\mathrm{A/NA}} \in \mathbb{R}^{(N_{\mathrm{spe}}+1)\times d}$), and scale ($\mathbf{S}_{\mathrm{A/NA}} \in \mathbb{R}^{(N_{\mathrm{spe}}+1)\times d}$) sets will be utilized for further comprehensive global correlation extraction on all tokens.

First, inspired by the efficient channel attention (ECA) [50] mechanism, a novel yet effective token feature attention (TFA) module is developed, which is dedicated to the weighted fusion of two token sets with different scale features. To do this, we first added up $\mathbf{S}_{\mathrm{A}}$ and $\mathbf{S}_{\mathrm{NA}}$ to get the initial multiscale feature, and then a depthwise 1-D convolution ($\mathrm{DWCon}v_{1\text{-}\mathrm{D}}$) operation followed by a sigmoid activation is adopted to generate the weight map, which is then multiplied back with the ($\mathbf{S}_{\mathrm{A}}+\mathbf{S}_{\mathrm{NA}}$) to obtain a new query set, i.e., $\mathbf{Q}_{\mathrm{A_NA}}$ as follows:

$$\mathbf{Q}_{\mathrm{A_NA}} = \mathrm{Sigmoid}\left(\mathrm{DWCon}v_{1\text{-}\mathrm{D}}\left(\mathbf{S}_A + \mathbf{S}_{\mathrm{NA}}\right)\right) \times \left(\mathbf{S}_A + \mathbf{S}_{\mathrm{NA}}\right) \tag{8}$$

where the kernel size of the $\mathrm{DWCon}v_{1\text{-}\mathrm{D}}(\cdot)$ is empirically set to 5 and $\mathbf{Q}_{\mathrm{A_NA}}$ holds the weighted multiscale abstract feature and also serves as the key set for the SIMHA. Then, the self-correlation map of each token from every single-scale perspective can be expressed as $\mathbf{R}_{\mathrm{speA}} = \mathrm{Softmax}((\mathbf{Q}_A\mathbf{K}_A^{\mathrm{T}}/(d)^{1/2}))$ and $\mathbf{R}_{\mathrm{speNA}}$=softmax($(\mathbf{Q}_{\mathrm{NA}}\mathbf{K}_{\mathrm{NA}}^{\mathrm{T}}/(d)^{1/2})$). Similarly, the correlation weight $\mathbf{R}_{\mathrm{A_NA}}$ within tokens can effectively be measured in a multiscale perspective by calculating from the softmax of the dot product of multiscale features $\mathbf{Q}_{\mathrm{A_NA}}$ using $\mathbf{R}_{\mathrm{A_NA}} = \mathrm{Softmax}((\mathbf{Q}_{\mathrm{A_NA}}\mathbf{Q}_{\mathrm{A_NA}}^{\mathrm{T}}/(d)^{1/2}))$. Next, the comprehensive spectral scale-aware global weighted representations $\mathbf{O}_{\mathrm{speA}}$ and $\mathbf{O}_{\mathrm{speNA}}$ of each head are calculated hierarchically, defined as

$$\mathbf{O}_{\mathrm{speA}} = \mathbf{R}_{\mathrm{speA}}\left(\mathbf{R}_{\mathrm{A_NA}}\mathbf{V}_{\mathrm{A}} + \mathbf{V}_{\mathrm{A}}\right) \tag{9a}$$

$$\mathbf{O}_{\mathrm{speNA}} = \mathbf{R}_{\mathrm{speNA}}\left(\mathbf{R}_{\mathrm{A_NA}}\mathbf{V}_{\mathrm{NA}} + \mathbf{V}_{\mathrm{NA}}\right). \tag{9b}$$

Similar to the conventional MHSA, the proposed SIMHA module repeats the aforementioned operation h times to calculate the attention scores of multiple heads. The calculated results of each head are, then, concatenated and mapped back to the size of the original dimension as $\mathrm{Concat}(\mathbf{O}^1_{\mathrm{speA}}, \mathbf{O}^2_{\mathrm{speA}}, \ldots, \mathbf{O}^h_{\mathrm{speA}})\boldsymbol{\omega}^O$ and $\mathrm{Concat}(\mathbf{O}^1_{\mathrm{speA}}, \mathbf{O}^2_{\mathrm{speA}}, \ldots, \mathbf{O}^h_{\mathrm{speA}})\boldsymbol{\omega}^O$, where $\boldsymbol{\omega}^O \in \mathbb{R}^{dh\times D}$ is the linear mapping function.

In this way, by considering two distinct input features $\mathbf{z}_{\mathrm{speA}}$ and $\mathbf{z}_{\mathrm{speNA}}$, the overall implementation process of SiTE can be expressed as follows:

$$\mathbf{F}'_l = \mathrm{SIMHA}\left(\mathrm{LN}\left(\mathbf{z}_{\mathrm{speA}}\right), \mathrm{LN}\left(\mathbf{z}_{\mathrm{speNA}}\right)\right) + \mathbf{F}_{l-1} \tag{10}$$

$$\mathbf{F}_l = \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{F}'_l\right)\right) + \mathbf{F}'_l \tag{11}$$

where the $\mathbf{F}_l = [\mathbf{F}^l_{\mathrm{speA}}, \mathbf{F}^l_{\mathrm{speNA}}]$ is the output of the SiTEs and l denotes the number of encoders. LN($\cdot$) represents the layer normalization operation, and MLP($\cdot$) is the multilayer perceptron (MLP), which consists of two linear layers with a Gaussian error linear unit (GELU) activation function between them. The proposed SiTEs are also employed to extract scale-aware global spatial features from the given multiscale spatial tokens, such as $\mathbf{z}_{\mathrm{spaL}}$ and $\mathbf{z}_{\mathrm{spaG}}$, in a manner analogous to the aforementioned scale-aware global spectral feature extraction method.

After the complete modeling of long-range relationships for the multiscale features, the class tokens (i.e., $\mathbf{T}_{\mathrm{speA}}$, $\mathbf{T}_{\mathrm{speNA}}$, $\mathbf{T}_{\mathrm{spaL}}$, and $\mathbf{T}_{\mathrm{spaG}}$) acquire highly discriminative class-related information generated by SIMHA. To obtain the high-level spectral and spatial information, we sum the multiscale class tokens that were obtained from the spectral and spatial branches, respectively, and it can be expressed as

$$\mathbf{T}_{\mathrm{spe}} = \mathbf{T}_{\mathrm{speA}} + \mathbf{T}_{\mathrm{speNA}} \tag{12}$$

$$\mathbf{T}_{\mathrm{spa}} = \mathbf{T}_{\mathrm{spaL}} + \mathbf{T}_{\mathrm{spaG}} \tag{13}$$

where $\mathbf{T}_{\mathrm{spe}}$ and $\mathbf{T}_{\mathrm{spa}}$ represent the final obtained class tokens of the spectral and spatial branches, respectively.

D. Stage 3: Spectral and Spatial Feature Fusions

It has been shown that the contribution of complementary spectral and spatial features is important to the

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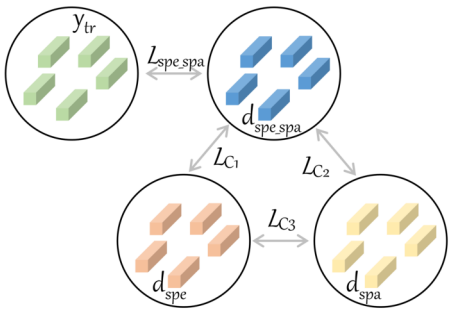


Fig. 6. Illustration of the proposed S^2C^2 loss function. The joint constraints of the loss functions L_{C1}, L_{C2}, and L_{C3} are employed to enhance the fusion efficiency of the spatial and spectral features. L_{spe_spa} is utilized to facilitate the consistency of the predictions generated by the final spectral–spatial fusion features with the labels, thereby controlling the direction of model learning.

decision-making process; it is also essential to have adaptability while combining these two different types of information. Furthermore, to keep the model simple, we avoid adding extra trainable parameters to the model at the feature fusion stage. Accordingly, a nonparametric fusion method, namely, S^2C^2 loss function (see Fig. 6) is proposed in order to achieve the decision-level information fusion. As the name suggested, we develop a new loss function to influence the training process of the model, thus strengthening the fusion of the two features. In particular, we constructed the spectral–spatial feature $\mathbf{T}_{spe_spa}$ by concatenating the obtained $\mathbf{T}_{spe}$ and $\mathbf{T}_{spa}$ using an empirically adopted concatenation operation before feeding them into the MLP layer. Consequently, the final decision vectors derived from different features are yielded, i.e., $\mathbf{d}_{spe} \in \mathbb{R}^{1\times C}$, $\mathbf{d}_{spa} \in \mathbb{R}^{1\times C}$, and $\mathbf{d}_{spe_spa} \in \mathbb{R}^{1\times C}$, where C represents the number of prediction categories.

Hereafter, in the training process, for a training sample $\mathbf{x}_{tr}$ with its true one-hot label $\mathbf{y}_{tr} \in \mathbb{R}^{1\times C}$, we can compute the main loss L_{spe_spa} for the proposed method, which can be mathematically expressed as

$$L_{spe_spa} = -\frac{1}{n}\sum_{i}^{n}\left[\mathbf{y}_{tr}^{i}\log\left(\mathbf{d}_{spe_spa}^{i}\right) + \left(1-\mathbf{y}_{tr}^{i}\right)\log\left(1-\mathbf{d}_{spe_spa}^{i}\right)\right] \quad (14)$$

where n is the number of training samples. Then, three constrained but auxiliary loss functions are suggested to regulate the fusion process in order to aid in the fusion of spectral and spatial features. These can be expressed as $L_{C1} = (1/n)\sum_{i=1}^{n}|\mathbf{d}_{spe} - \mathbf{d}_{spe_spa}|^2$, $L_{C2} = (1/n)\sum_{i=1}^{n}|\mathbf{d}_{spa} - \mathbf{d}_{spe_spa}|^2$, and $L_{C3} = (1/n)\sum_{i=1}^{n}|\mathbf{d}_{spa} - \mathbf{d}_{spe}|^2$, where L_{C1}, L_{C2}, and L_{C3} are the consistency constraint loss functions. Finally, the proposed S^2C^2 loss function $L_{S^2C^2}$ can be formed as follows:

$$L_S^2C^2 = L_{spe_spa} + \frac{1}{3}L_{C1} + \frac{1}{3}L_{C2} + \frac{1}{3}L_{C3}. \quad (15)$$

Building upon the aforementioned exposition, we can encapsulate the merits bestowed by the congruity between constrained spectral features, spatial features, and spectral–spatial fusion feature prediction outcomes as follows: 1) dynamically harmonizing the influences of spectral and spatial features on the ultimate decision and 2) harmonizing the subdecision with the primary decision fosters a balanced convergence of the dual-branch network, thereby enhancing fusion efficacy.

IV. Experiments

In this section, we conduct a comprehensive assessment and discuss the efficacy of the proposed classification method across four widely used benchmarks. First, we offer a succinct overview of these datasets, followed by a detailed description of the experimental implementation setup. Then, we conduct a comprehensive analysis and discussion of the proposed SiT, including parameter sensitivity and an ablation analysis. Finally, substantial comparative studies provide a quantitative and qualitative evaluations of the proposed method.

A. Data Description

To fully validate the reliability of the proposed method, we perform experiments over the four datasets leveraging the classification tests. The first dataset is *Indian Pines* (IP), comprising of 145 × 145 pixels with 200 effective spectral bands. The second dataset employed in this article is the *Pavia University* (PU), which encompasses a spatial size of 610 × 340 pixels along with 103 high-quality spectral bands. The third adopted dataset is the *University of Houston2013* (UH),[1], [2] specifically provided for the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest, which comprises a spectral dimension of 144 spectral channels and consists of 349 × 1905 pixels. The fourth dataset is the *HyRANK-Dinoi* (HD)[3] [51], which possesses a spatial size of 250 × 1376 pixels along with 176 spectral channels.

Notably, to enable a fair experimental comparison, we employ the spatially disjoint training and test sets offered by Paoletti et al. [52] for all experiments, which are widely adopted in numerous recent works [36], [42], [44], in the quest for credible experimental results. Motivated by the sampling approach of the TAIGA dataset [53], we constructed a more rigorous spatially disjoint training/testing split on the fourth dataset, leveraging the original labeled areas of the HD scenes. The pseudo-color images of these four HS scenes together with the corresponding distribution of training and test samples are depicted in Fig. 7. Moreover, the specific number of training and test samples is reported in Table I.

B. Experimental Setup

1) Assessment Criteria: In this study, all classification results of the compared and the proposed methods were comprehensively evaluated using both quantitative and qualitative criteria. The quantitative assessment will encompass several metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient, which provides objective for the measurements of the model's performance and its ability to accurately classify the different land covers. The qualitative evaluation will be conducted by examining the generated classification maps, which visually illustrate the spatial distribution of the predicted classes.

[1]http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

[2]The cloud-free version was provided by Prof. N. Yokoya from the University of Tokyo and RIKEN AIP.

[3]https://www.isprs.org/society/si/SI-2017

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR FOUR DATASETS

No.	Indian Pines			Pavia University			University of Huston 2013			HyRANK-Dinoi		
	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test
1	Alfalfa	29	25	Asphalt	548	6304	Healthy Grass	198	1053	Urban Fabric	234	1028
2	Corn-notill	762	675	Meadows	540	18146	Stressed Grass	190	1064	Mining Sites	100	104
3	Corn-mintill	435	404	Gravel	392	1815	Synthetic Grass	192	505	Dryland Arable	105	509
4	Corn	146	99	Trees	524	2912	Tree	188	1056	Fruit Trees	29	121
5	Grass-pasture	232	274	Metal Sheets	265	1113	Soil	186	1056	Olive Groves	322	1446
6	Grass-trees	394	354	Bare Soil	532	4572	Water	182	143	Coniferous Forest	29	332
7	Grass-past-m	16	2	Bitumen	375	981	Residential	196	1072	Dense Sclerophyll Cover	523	4512
8	Hay-windowed	235	250	Briks	514	3364	Commercial	191	1053	Sparce Sclerophyll Cover	560	5814
9	Oats	10	10	Shadows	231	795	Road	193	1059	Sparcely Vegetated Areas	347	1407
10	Soybean-notill	470	503				Highway	191	1036	Rocks and Sand	78	387
11	Soybean-mintill	1424	1065				Railway	181	1054	Water	447	1165
12	Soybean-clean	328	282				Parking Lot1	192	1041	Coastal Water	73	325
13	Wheat	132	80				Parking Lot2	184	285			
14	Woods	728	545				Tennis Court	181	247			
15	Build-g-t-d	291	99				Running Track	187	473			
16	Stone-steel-t	57	44									
	Total	5689	4721	Total	3921	40002	Total	2832	12197	Total	2847	17150

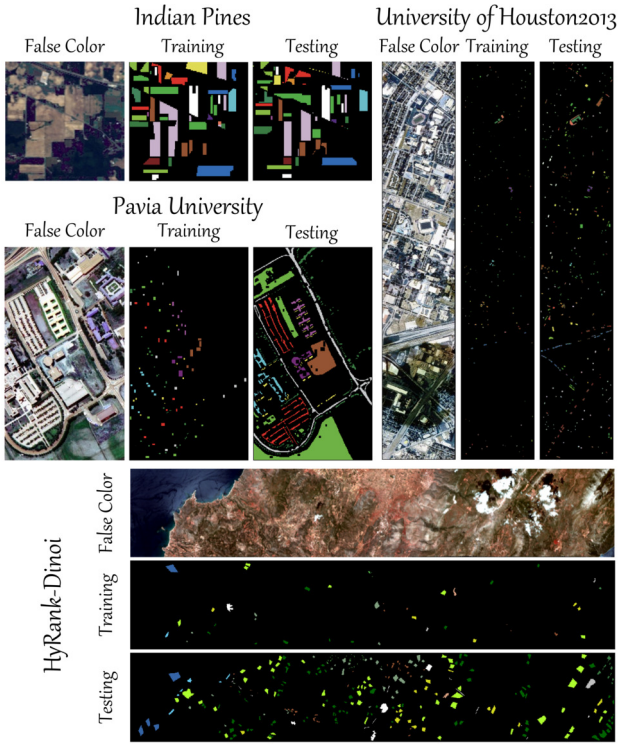


Fig. 7. False-color images and distributions of training/testing samples for the four datasets.

2) Comparison With State-of-the-Art Baseline Method: To provide a comprehensive and meaningful evaluation of the performance of the proposed model, we have selected 11 representative methods for comparison, including spectral–spatial residual network (SSRN) [30], HybridSN [54], double-branch dual-attention (DBDA) network [55], attention-based adaptive spectral–spatial kernel (A2S2K) network [56], ViT [25], SpectralFormer (SF) [36], SSFTT [38], BS2T [42], morphological transformer (morphFormer) [44], CMTFNet [57], and RSSFormer [58], respectively. Notably, the ViT directly utilizes transformer encoders for feature extraction from HSI images, with the number of encoders and input window size adjusted to optimize classification performance. For all other comparison methods, we have reproduced them following the original article and its official code.

3) Implementation Details: All experiments, including the proposed SiT and other comparative methods, were conducted using the PyTorch 1.10.0 framework embedded on a workstation with an RTX 3090 graphics processing unit. Additionally, we choose the Adam optimizer with an initial learning rate of $5e^{-3}$ to optimize all trainable parameters of the method in a batch size of 64. To facilitate effective model fitting, we employed a learning rate decay strategy during training the model. In our experiments, the total number of training epochs was set to 150, 150, 100, and 100 for four datasets, respectively. It can be noted that we multiplied the learning rate by 0.9 for every tenth of completed training epochs.

C. Analysis of Hyperparameters

As mentioned before, the amount of information the model obtains depends on the input size (i.e., w_1, w_2, and P) of the proposed SiT model, which inevitably impacts the classification outcomes. Furthermore, another two hyperparameters associated with the structure of the model, i.e., the number of SiTEs (l) and heads (h), which may have an impact on the quality of feature extraction, will be examined. As a result, we conducted comprehensive experiments and analysis in order to pursue the optimal parameter settings.

1) Impact of w_2 and P on Classification Performance: As aforementioned, these two hyperparameters control the volume of spatial information the model can assimilate, influencing whether the spatial feature extraction branch can yield discriminative spatial features appropriate for the final classification decision. Therefore, we jointly discuss the classification performance achieved in terms of OA of the proposed SiT when w_2 is in the range of $\{9, 10, \ldots, 21\}$ and P is in the range of $\{5, 10, \ldots, 30\}$, respectively, for all four datasets jointly. Notably, the initial setting of the other three hyperparameters (i.e., w_1, h, and l) is set to 1, 4, and 1, respectively. The collective effect of the spatial patch size and the number of retained

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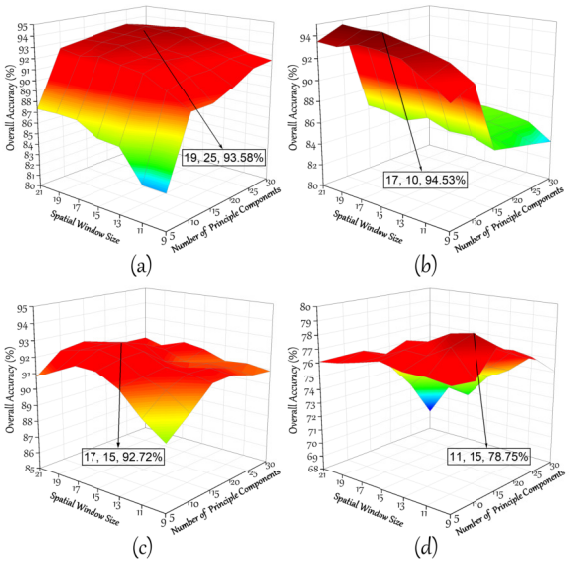


Fig. 8. Impact of spatial window size (w_2) and the number of principal components (P) on classification performance across four datasets. (a) IP dataset. (b) PU dataset. (c) UH dataset. (d) HD dataset.

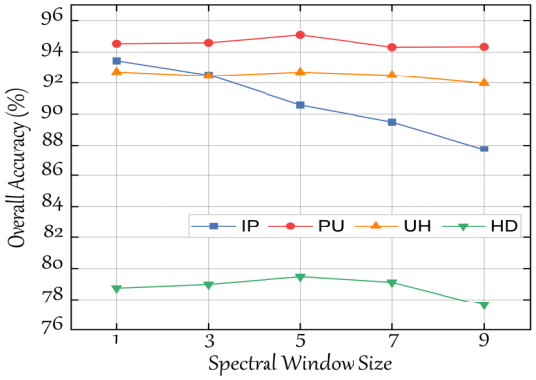


Fig. 9. Impact of the spectral window size (w_1) on classification performance.

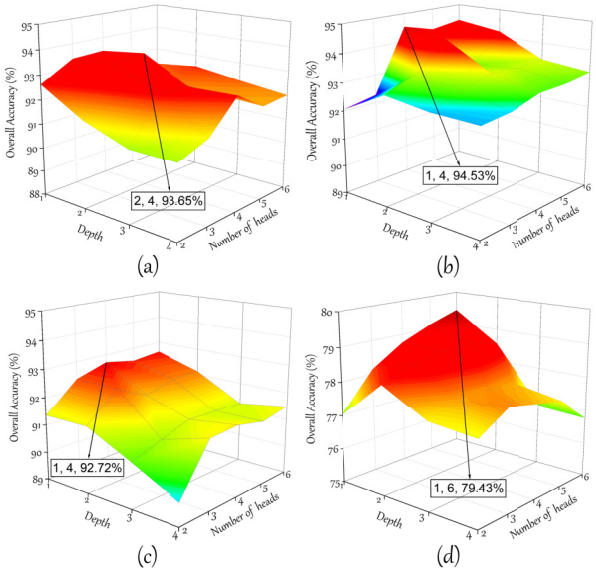


Fig. 10. Impact of the number of SiTE (l) and heads (h) on classification performance across four datasets. (a) IP dataset. (b) PU dataset. (c) UH dataset. (d) HD dataset.

principal components results in regular fluctuations/variation in the classification performance of the proposed SiT method, as visualized in Fig. 8. These variations show a variety of tendencies depending on the datasets being analyzed. For the IP dataset, the optimal point for classification performance is located farther into the process due to the higher spatial homogeneity of land covers and the large number of spectral bands. Additionally, the simplicity of the land cover in this scenario enables that once sufficient spatial information is provided, further increases in w_2 yield no significant improvement in classification performance. Because of the complex distribution of urban and wilderness land covers, the PU, UH and HD datasets using excessively large spatial input patches may lead to the inclusion of too many outlier features, which could affect classification performance. In particular, for the HD dataset, which lacks human management, w_2 should be set to smaller values. Moreover, preserving an excessive number of spectral principal components can negatively affect classification accuracy because these three datasets have very few bands, particularly in the case of the PU dataset. The experimental results indicate that the optimal values of (w_2, P) for the four datasets are (19, 25), (17, 10), (17, 15), and (11, 15), respectively. Since the accuracy of the IP dataset at $w_2 = 17$ is not much different from that at $w_2 = 19$, we set w_2 to 17 in subsequent experiments on the IP dataset.

2) Impact of w_1 on Classification Performance: The local spectral information that is considered to be the same material depends on the input patch size of the spectral feature extraction path. Therefore, we further investigated the impact of the value of w_1 on model performance. As w_1 progressively increases, the evolving trend of the classification precisions (in terms of OA) for four datasets is shown in Fig. 9. For the IP dataset, the classification accuracy significantly decreases as w_1 increases. Notably, the incorporation of more spectral features, relative to other datasets, led to a decline in classification accuracy on the IP dataset, indicating severe spectral variability within its intraneighborhood spectra. In contrast, the classification performance of SiT on the PU, UH, and HD datasets is less influenced by the variation in w_1. The classification performance of SiT on the PU and HD datasets even tends to rise within a certain range of w_1 due to more information being provided. Moreover, unlike the PU dataset, increasing w_1 gradually decreases SiT's classification accuracy on the UH and HD datasets, as their more complex land-cover distributions are more vulnerable to the inclusion of alien spectral information. Eventually, we uniformly set w_1 to 1 for all four datasets to facilitate the practical application.

3) Impact of l and h on Classification Performance: These two structural hyperparameters have an impact on the model's ability to extract features to some extent, further reflecting the influence of the modification in model structure on the ability to fit the data. Therefore, it is essential to search for the appropriate parameter setting. Similarly, we investigate the sensitivity of these two hyperparameters jointly on four datasets. Fig. 10 illustrates the trends in classification

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TABLE II

ABLATION ANALYSIS OF THE PROPOSED SiT WITH DIFFERENT MODULES (OR STRATEGIES) ON FOUR DATASETS IN TERMS OF OA

Networks	Ablation Step		OA (%)			
	SiTE	S^2C^2 Loss	IP	PU	UH	HD
1	✗	✗	84.78	91.39	88.03	70.28
2	✓	✗	88.03	93.47	89.17	73.66
3	✗	✓	91.76	92.85	90.93	76.96
4	✓	✓	**93.44**	**94.53**	**92.72**	**78.75**

performance affected by these two hyperparameters. It can be observed that the affected trends reflected in all datasets are similar. Specifically, the vanishing gradient problem and the over-parameterization problem caused by the increase in l and h render the model incapable of fitting the data well, thereby causing the performance to trend downward. It is worth noting that the rate of decrease in accuracy for the IP and PU datasets becomes slow while the depth increases to a certain level. This is because these two datasets provide more training samples than the other datasets, providing more comprehensive features for model learning. Moreover, an insufficient number of heads might limit performance by reducing the comprehensiveness of features that can be extracted. We have indicated in Fig. 10 the specific parameter settings for SiT that achieve the best performance on each dataset. Finally, we have subsequently set both h and l to 4 and 1 for the sake of practical application.

D. Ablation Experiment

We carried out extensive studies to investigate the gains of classification performance achieved by the two proposed strategies, i.e., SiTE and S^2C^2 loss, respectively, in terms of OAs by gradually embedding them in a stepwise fashion on the four datasets. The classification accuracy of different ablation studies is detailed in Table II. It is worth noting that the parameter settings are kept similar to those in Section IV-C. Furthermore, the configuration of the various networks involved in the ablation experiments is described in the following for the purpose of clarity.

1) For Network_1, it only adopts two individual vanilla transformer encoders to process multiscale features and does not utilize the S^2C^2 loss function.
2) For Network_2, in contrast to Network_1, we replace the original transformer encoder with the proposed SiTE.
3) For Network_3, building on the foundation of Network_1, incorporate the use of the S^2C^2 loss function to facilitate the fusion of spectral and spatial features.
4) For Network_4, it concerns the approach presented in this article, which integrates both SiTE and S^2C^2 loss functions.

As can be observed, Network_2, which considers the comprehensive multiscale features, achieves better classification than Network_1 across all datasets, which, to some degree, suggests that implementing global attention inside the transformer encoder, especially from a local multiscale perspective, is feasible for improving the HSIC classification accuracy. Moreover, Network_3 exhibits superior advantages when compared to Network_1 (beyond around 3.25%, 1.51%, 2.9%, and 3.38% OAs, respectively) when the S^2C^2 loss function is incorporated. Interestingly, it can be observed that the S^2C^2 loss function brings a higher gain to the classification performance on the IP, UH, and HD datasets than on the PU dataset. Particularly, the S^2C^2 loss function has an extremely prominent effect on the classification accuracy of the IP and HD datasets. This phenomenon arises from differences in spatial and spectral resolution among the IP, UH, and HD datasets, resulting in an uneven distribution of the diagnostic characteristics of land cover across spatial and spectral dimensions. Consequently, it becomes essential to extract both features during the training process in an adaptive way. Certainly, additional underlying reasons for this phenomenon exist, warranting further exploration in future studies. However, it is essential to acknowledge that leveraging the S^2C^2 loss function to guide model training yields more satisfactory classification performance than merely fusing the two features directly. Additionally, even more significant performance gains are obtained by combining SiTE with S^2C^2 loss function in a synergistic fashion. To more intuitively depict the performance gains attributed to SiTE and S^2C^2 loss function, we employ t-stochastic neighbor embedding (T-SNE) on all datasets to show the feature mapping derived from a low-dimensional space of these four networks for visualization of feature separability. Alongside Fig. 11, we can draw conclusions akin to those that were previously addressed. The fact that the S^2C^2 loss function has a stronger impact on improving interclass separability is even more notable.

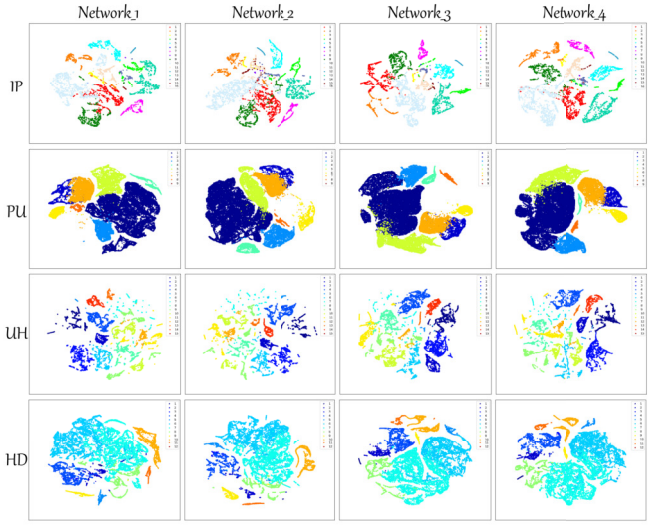


Fig. 11. Feature embedding visualizations of ablation experiments on four datasets.

E. Qualitative Evaluation

Tables III–VI report the quantitative summary of the accuracy for each category in terms of OA, AA, and Kappa values obtained by the proposed method and the aforementioned comparative methods over the IP, PU, UH, and HD datasets, respectively. Furthermore, the obtained best results are highlighted in bold.

As can be observed that the CNN- and transformer-based comparative methods show varied classification performance

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TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT STATE-OF-THE-ART CLASSIFICATION ALGORITHMS ON IP DATASET. THE BOLD ENTITIES INDICATE THE BEST ONE

Class.	CNN-based Network				Transformer-based Network							
	SSRN	HybridSN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	morphFormer	CMTFNet	RSSFormer	Proposed
1	96.01	95.65	92.38	91.76	20.39	56.47	**100**	**100**	91.66	65.21	79.32	**100**
2	90.81	75.7	74.96	79.26	75.61	79.85	87.41	80.74	79.26	79.11	93.33	**92.59**
3	75.74	85.15	91.99	82.43	76.73	57.18	76.24	75.74	86.39	79.7	80.69	**93.56**
4	66.67	26.61	56.57	57.58	51.52	43.43	54.55	**91.92**	49.49	47.47	80.81	77.78
5	90.88	68.61	91.61	88.32	62.04	81.75	90.15	92.34	90.88	88.32	**95.99**	90.88
6	96.61	99.15	97.46	98.26	96.33	95.48	94.07	92.37	94.42	**99.72**	89.38	99.44
7	0	0	0	0	0	0	0	0	0	0	0	0
8	**100**	**100**	**100**	**100**	**100**	94.8	**100**	**100**	97.2	97.99	**100**	**100**
9	40	60	0	80	40	**100**	60	30	**100**	70	80	90
10	80.32	80.52	96.82	84.69	84.69	89.99	87.87	**96.42**	92.84	92.64	74.55	85.29
11	**96.06**	82.44	91.83	93.52	85.92	86.38	86.85	90.05	84.23	87.32	92.58	93.62
12	91.84	52.48	**97.87**	84.75	72.34	62.06	85.46	97.87	85.11	66.67	71.99	93.97
13	96.25	97.5	96.25	91.25	86.25	**100**	93.75	96.03	85.49	98.75	78.75	**98.75**
14	99.27	98.35	**100**	97.43	97.06	95.05	97.06	98.97	98.96	96.33	98.9	99.45
15	72.73	64.65	62.63	56.57	54.55	80.81	17.17	65.66	11.11	75.76	10.1	**93.94**
16	86.36	86.36	97.73	79.55	90.91	97.73	72.73	79.55	72.73	**100**	77.27	86.36
OA (%)	90.59	81.65	90.61	88.11	82.38	82.93	86.47	90.02	85.93	86.52	87.05	**93.44**
AA (%)	79.97	73.32	77.95	79.09	68.39	76.31	75.21	80.47	76.23	77.81	75.22	**87.22**
Kappa	0.8926	0.7919	0.8932	0.8645	0.7997	0.8051	0.8466	88.67	84.04	0.8468	0.8528	**0.9265**

TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT STATE-OF-THE-ART CLASSIFICATION ALGORITHMS ON PU DATASET. THE BOLD ENTITIES INDICATE THE BEST ONE

Class.	CNN-based Network				Transformer-based Network							
	SSRN	HybridSN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	morphFormer	CMTFNet	RSSFormer	Proposed
1	92.96	93.39	90.77	92.34	76.06	90.7	90.39	91.36	**93.56**	87.12	88.61	92.66
2	76.11	73.85	78.32	82.82	84.66	98.07	71.92	71 .47	78.62	93.82	98.47	**98.77**
3	81.21	77.19	**78.24**	69.75	47.11	71.18	82.70	75.67	74.93	73.99	74.82	78.14
4	**99.86**	70.57	92.82	98.70	97.18	94.61	87.40	92.32	79.46	96.29	95.67	99.55
5	**100.0**	99.28	98.02	97.66	99.01.	**100.0**	99.10	93.23	98.74	99.55	99.46	**100.0**
6	98.80	99.85	99.85	95.34	59.01	73.63	**100.0**	93.21	98.69	80.07	63.47	87.27
7	**100.0**	95.21	99.69	90.72	62.49	80.3	99.59	92.12	96.64	98.06	97.45	96.74
8	98.22	97.92	99.44	98.48	91.97	82.65	95.33	93.94	98.54	98.75	97.65	**99.71**
9	99.50	93.84	97.1 1	97.86	90.82	99.89	99.12	92.14	93.58	95.97	96.35	**100.0**
OA (%)	86.89	83.61	87.01	88.53	80.17	90.71	83.59	87.09	86.14	91.19	91.49	**94.53**
AA (%)	94.07	89.01	92.69	91.52	78.70	87.89	91.27	88.38	90.31	91.51	90.22	**94.75**
Kappa	0.8315	0.7876	0.8322	0.8502	0.7360	0.8755	0.7906	0.8587	0.8205	0.8815	0.8839	**0.9279**

TABLE V

QUANTITATIVE COMPARISON OF DIFFERENT STATE-OF-THE-ART CLASSIFICATION ALGORITHMS ON UH DATASET. THE BOLD ENTITIES INDICATE THE BEST ONE

Class.	CNN-based Network				Transformer-based Network							
	SSRN	HybridSN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	morphFormer	CMTFNet	RSSFormer	Proposed
1	82.34	80.01	82.72	84.05	82.53	**94.58**	82.81	81.57	82.53	82.91	82.91	86.89
2	95.68	82.77	98.97	99.25	96.15	97.71	88.16	82.34	**100.0**	84.68	90.7	99.15
3	99.8	81.78	99.80	**100.0**	87.33	93.31	95.25	93.21	96.24	98.61	98.22	**100.0**
4	**99.15**	75.66	98.11	99.15	97.06	97.32	85.8	92.38	96.97	88.54	89.96	97.25
5	**100.0**	98.11	**100.0**	**100.0**	99.34	99.28	**100.0**	99.17	98.58	**100.0**	**100.0**	99.91
6	95.8	90.21	95.80	95.80	93.01	73.04	**100.0**	86.21	**100.0**	95.8	97.2	**100.0**
7	88.43	69.59	**93.38**	91.70	89.46	61.31	75.19	81.42	92.91	80.88	87.59	87.41
8	74.36	83.86	65.24	72.65	78.06	73.57	47.67	65.42	81.21	86.13	75.88	**94.11**
9	85.93	66.38	81.78	90.27	76.96	84.56	82.53	71.01	86.59	81.96	86.87	**91.03**
10	72.01	76.69	**81.66**	71.14	55.69	59.82	55.02	67.39	54.05	95.37	89.19	81.27
11	98.39	88.33	96.68	98.77	68.60	92.81	88.14	90.07	95.35	**99.53**	98.67	99.05
12	**92.89**	82.61	81.94	64.36	68.97	75.49	92.7	81.39	88.95	90.49	89.43	85.98
13	85.26	78.61	85.61	92.63	89.47	67.61	**94.39**	74.43	84.56	75.79	81.05	87.37
14	**100.0**	**100.0**	**100.0**	**100.0**	99.60	98.45	**100.0**	**100.0**	99.59	**100.0**	**100.0**	**100.0**
15	**100.0**	99.12	**100.0**	**100.0**	**95.56**	98.99	**100.0**	99.05	**100.0**	**100.0**	**100.0**	**100.0**
OA (%)	90.04	81.92	89.31	88.92	82.86	84.46	82.24	87.69	88.92	89.83	90.05	**92.72**
AA (%)	91.33	83.58	90.77	90.65	85.18	84.52	85.84	84.34	90.5	90.71	91.17	**93.96**
Kappa	0.8468	0.8037	0.8921	0.8776	0.8141	0.8319	0.8084	0.8273	0.8797	0.8896	0.8922	**0.9238**

on four different datasets. Specifically, the CNN-based methods achieve relatively superior performance in most cases than that of pure transformer-based comparative methods (i.e., ViT and SF) even with the complex land-cover distribution (e.g., UH and HD datasets). This observed phenomenon can potentially be attributed to the lack of inductive bias of transformer-based architecture. Although the SSFTT and BS2T adopt feature maps derived by a convolution operation as the input of the transformer encoders, the classification performance is limited since the tokens obtained from fixed-sized convolutional receptive fields do not provide a comprehensive description of the land covers. In contrast, the morphFormer obtains feature maps containing morphological shape information through the application of widely used morphological

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TABLE VI

QUANTITATIVE COMPARISON OF DIFFERENT STATE-OF-THE-ART CLASSIFICATION ALGORITHMS ON HD DATASET. THE BOLD ENTITIES INDICATE THE BEST ONE

Class.	CNN-based Network				Transformer-based Network							
	SSRN	HybridSN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	morphFormer	CMTFNet	RSSFormer	Proposed
1	20.19	56.03	29.86	46.3	30.54	61.96	43.19	47.08	**65.18**	63.91	47.06	51.56
2	95.19	75.96	94.23	97.12	96.47	90.38	77.88	**97.94**	81.73	95.19	97.49	95.19
3	27.5	27.11	83.12	**84.48**	27.69	37.52	81.73	16.01	11.79	63.46	68.57	79.17
4	82.64	62.81	**97.52**	93.39	63.03	80.99	90.91	**97.52**	79.34	97.52	89.26	93.39
5	62.97	67.08	27.52	53.39	**74.27**	71.65	21.72	41.49	54.75	19.25	55.19	58.58
6	98.32	79.82	**99.34**	98.19	62.35	98.19	98.8	98.27	91.57	98.42	98.96	96.99
7	91.62	86.79	91.11	92.05	93.41	92.75	89.21	91.42	90.58	87.88	88.08	**94.66**
8	59.86	58.81	63.19	59.37	52.15	51.96	62.81	63.07	66.55	68.75	62.69	**67.73**
9	38.02	35.82	73.7	65.67	53.23	76.12	51.74	82.26	**90.48**	70.36	62.09	88.56
10	98.97	98.71	98.97	**99.74**	53.49	98.97	57.88	98.97	98.97	98.97	98.97	98.97
11	**100**	90.73	81.19	49.79	**100**	37.34	**100**	**100**	46.78	**100**	**100**	**100**
12	88.62	90.77	28.66	68.95	74.77	28.05	89.85	35.38	65.85	93.91	**94.46**	81.54
OA (%)	68.55	68.08	69.49	69.83	67.18	67.46	68.68	71.69	72.18	72.64	72.78	**78.75**
AA (%)	71.99	69.21	72.36	75.7	65.11	68.82	72.14	72.45	70.29	79.8	80.23	**83.86**
Kappa	0.6061	0.6075	0.6231	0.6281	0.5947	0.602	0.6109	0.6478	0.6554	0.6592	0.6635	**0.7401**

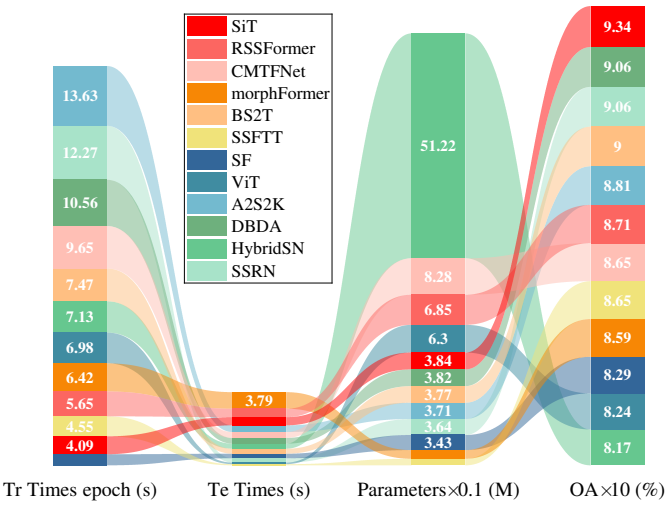


Fig. 12. Model complexity analysis of different DL-based methods on IP dataset.

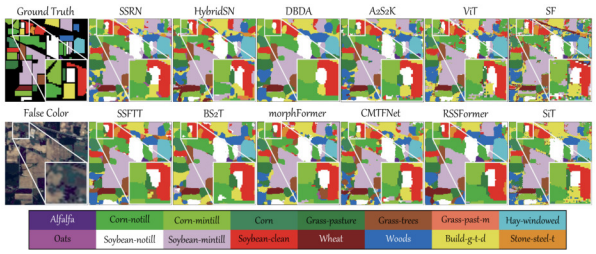


Fig. 13. Visualization of classification result compared with existing HSIC methods obtained on the IP dataset. The best classification map is suggested to be evaluated by jointly comparing the ground-truth image with the false-color image.

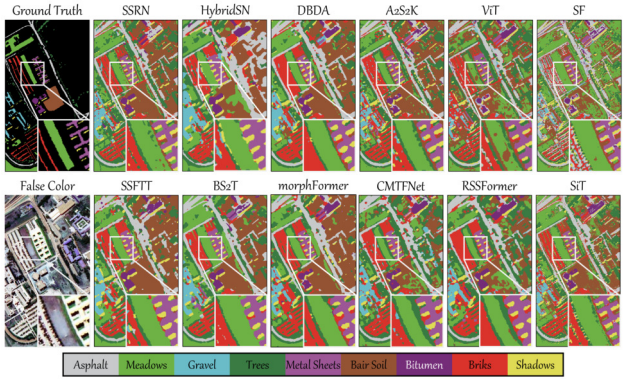


Fig. 14. Visualization of ground-truth map and classification result obtained by different methods on the PU dataset. The best classification map is suggested to be evaluated by jointly comparing the ground-truth image with the false-color image.

erosion and dilation operations before the MHSA mechanism, yielding relatively superior outcomes. With similar motivation, CMTFNet and RSSFormer have also attained appreciable performance by extracting spatial context features in the front and back segments of the original transformer structure, respectively. However, morphFormer, CMTFNet, and RSSFormer's preference for extracting spatial context relations while failing specifically to extract spectral features causes them to perform poorly in recognizing elongated structural ground cover, such as Category 1 Asphalt in the PU scenario and Category 9 Road in the UH scenario. On the other hand, the four SOTA CNN-based networks suffer from classification performance due to their inability to capture long-distance correlation information.

Unlike the transformer-based models described above, our proposed SiT not only fully considers the extraction of multiscale spatial and spectral features, but also further explores the global correlation from the perspective of multiscale features. Remarkably, our proposed SiT provides outstanding performance over all four datasets, demonstrating excellent generalization ability. More specifically, the proposed SiT achieves OA of 93.44%, 94.53%, 92.72%, and 78.75% with a gain of 2.83%, 3.04%, 2.67%, and 5.97% over the best results obtained by other methods on the IP, PU, UH, and HD datasets, respectively. It is worth noting that in the classification results for the IP dataset, class 7 suffers from overfitting, preventing all methods from effectively recognizing test set samples, despite having a significantly higher number of training samples. In addition, the proportion of the training set and the difficulty of extracting diagnostic features varied across land-cover categories, leading to differences in classification performance. For instance, in the HU and HD datasets, some categories achieve 100% accuracy, whereas others fall below 90%. However, our method is more competitive in balancing multiclass recognition performance. Furthermore, when

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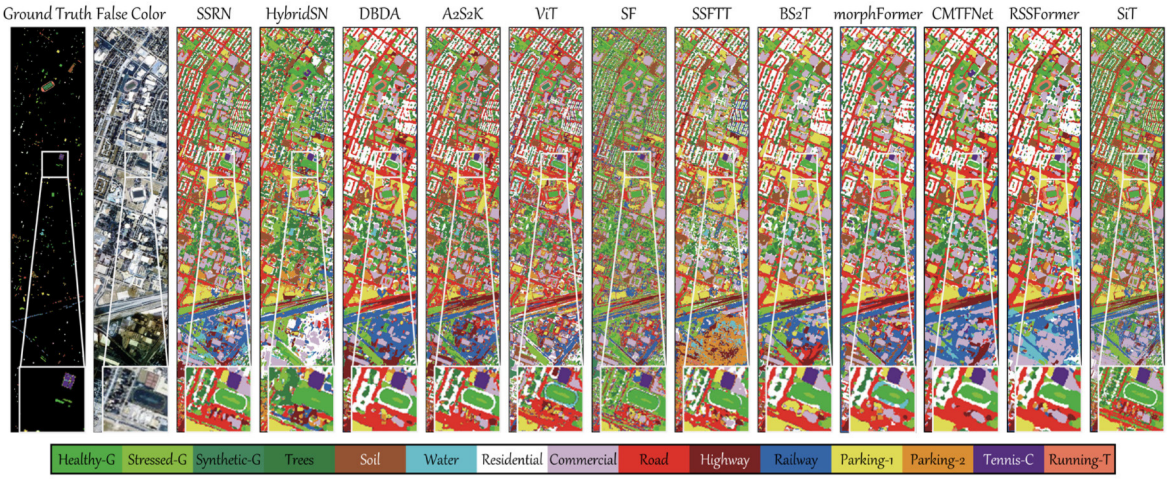


Fig. 15. Visualization of ground-truth map and classification result obtained by different methods on the UH dataset. The best classification map is suggested to be evaluated by jointly comparing the ground-truth image with the false-color image.

further considering its classification performance across multiple datasets, our approach exhibits greater robustness compared to other methods.

We further investigated the computational complexity of all DL-based methods over the IP dataset in terms of trainable parameters, testing time, and training time per epoch, respectively. As illustrated in Fig. 12, although our approach achieves superior performance in terms of classification accuracy, our network is intentionally designed to use a wider structure to ensure more comprehensive features are considered during network training. However, this design results in our proposed method lacking a significant advantage over other methods in terms of computational complexity. Naturally, investigating methods to reduce the complexity of our model is one of our upcoming tasks. Another noticeable point is that even with more parameters, the original ViT achieves the worst classification accuracy while using more parameters, indicating its low feature extraction efficiency.

F. Visual Comparison

To provide a more intuitive yet visual description of the classification performance of the various methods, we generated the actual classification maps for high-level visual analysis. Visualizations of the classification results for all the methods over the IP, PU, UH, and HD datasets are shown in Figs. 13–16, respectively. Before analyzing the classification result maps obtained by different methods, we must emphasize that the subjective evaluation of the land-cover classification effect should not rely solely on the provided ground-truth maps. It should also be analyzed in conjunction with the actual false-color images. This is because ground-truth maps for remote sensing images are typically created using robust, regionally consistent land-cover labeling, which makes it difficult to achieve highly granular and accurate labeling. Consequently, the land-cover classification results in the classification map should be consistent not only with the ground-truth map but also with the actual image's land-cover distribution. Consequently, when evaluating classification performance using the classification map, it should be compared not only with the ground truth but also with the actual land-cover characteristics.

Overall, most classification methods generated clean classification maps for all four datasets, except for ViT and SF. This is in part attributable to the fact that they do not use a convolutional bias, suggesting that it is difficult to obtain a sufficiently satisfactory classification map using only the transformer. However, it is crucial to emphasize that cleaner classification result maps do not necessarily imply more realistic classification results. Due to an over-reliance on spatial features for pixel classification, most methods resulted in significant interclass intrusion phenomena in the classification result maps relative to the actual land-cover distribution. One manifestation of this phenomenon is that the land-cover boundaries in the classification result map do not correspond to the actual situation, such as the boundary between "Grass-trees" and "Corn-notill" in Fig. 13. It can be seen that our proposed SiT provides the most accurate judgment of this boundary. Another manifestation of interclass intrusion is the incorrect identification of two or more neighboring land covers as the same class. This issue is especially pronounced in complex urban scenarios. In such scenarios, large and small land covers are usually adjacent, which can easily cause the model to overlook the presence of the smaller land cover during the recognition process. For example, in Fig. 14, to the left of the zoomed-in view area is a parking lot with planted trees. The ground-truth map, however, only labels the road in the parking lot, leading some methods (e.g., DBDA and RSSFormer) to identify the entire area as a road. In contrast, our method accurately identifies not only the road but also the tree (dark green) and the shadow (yellow), which are not labeled but actually exist. Moreover, in complex urban scenes (see Fig. 15), our method achieves excellent intraclass consistency prediction (playgrounds) and maintains clear interclass boundaries (roads). Similarly, in Fig. 16, our proposed method produces a classification result that more accurately aligns with the actual coastline. From the above discussion,

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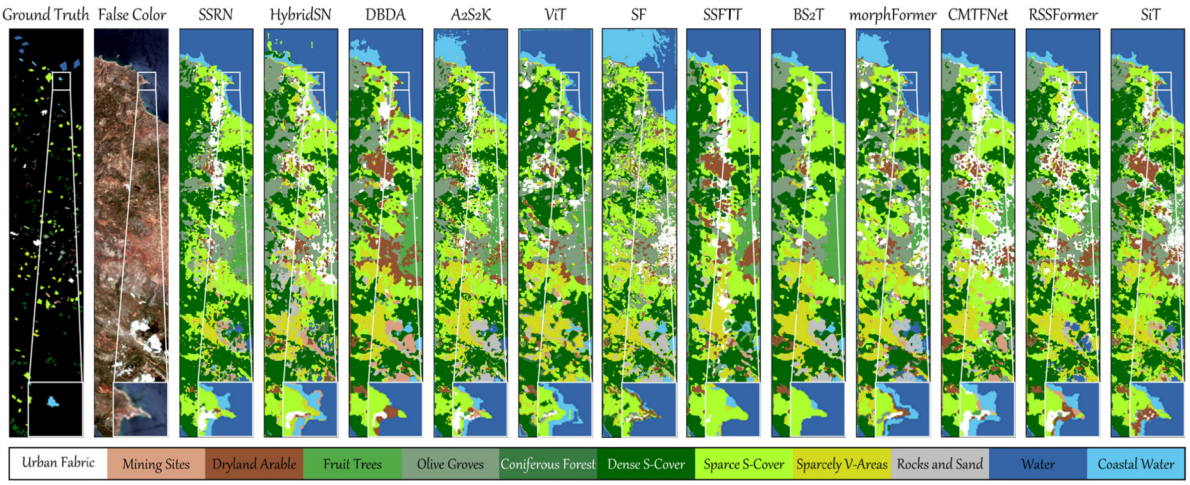


Fig. 16. Visualization of ground-truth map and classification result obtained by different methods on the HD dataset. The best classification map is suggested to be evaluated by jointly comparing the ground-truth image with the false-color image.

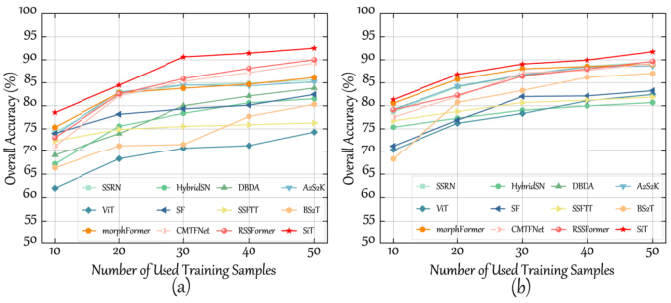


Fig. 17. Classification performance (OA) obtained by different methods with varying number of training samples on (a) PU dataset and (b) UH dataset.

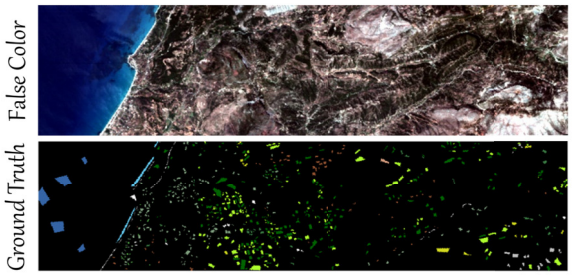


Fig. 18. Pseudo-color image and ground-truth map of HL dataset.

we believe the key to achieving high intraclass consistency and mitigating interclass intrusion in classification maps lies in how the model balances spatial and spectral features in the classification decision. Our proposed model, SiT, not only effectively extracts spectral and spatial features with high discriminative power but also adaptively fuses these features by introducing the S^2C^2 loss function to constrain the consistent identification of spatial and spectral features, enhancing the model's finesse and robustness for land-cover classification. In summary, the classification visualization results of SiT across the four datasets demonstrate superior classification performance compared to other models.

TABLE VII

NUMBER OF TRAINING AND TESTING SAMPLES ON THE HYRANK DATASET

No.	Class	HD (Train)	HL (Test)
1	Urban Fabric	206	288
2	Mining Sites	54	67
3	Dryland Arable	426	542
4	Fruit Trees	79	79
5	Olive Groves	1107	1401
6	Coniferous Forest	422	500
7	Dense Sclerophyll Cover	2996	3793
8	Sparce Sclerophyll Cover	2361	2803
9	Sparcely Vegetated Areas	399	404
10	Rocks and Sand	453	487
11	Water	1393	1393
12	Coastal Water	421	451
	Total	19997	12208

G. Impact of the Number of Training Samples

To accommodate the challenge of acquiring labeled samples in real-world application scenarios, we further extend our investigation to explore the impact of varying proportions of samples from the given training set on the classification tasks. The samples of the standard training set taken from the GRSS DASE website show a notably high intraclass similarity, as indicated by previous findings [59]. Consequently, we opt for a significantly smaller number of the sample size for the training set to better assess the performance of various models under the available limited samples. Specifically, varying numbers of training samples are randomly selected from the PU and UH training datasets, ranging from 10 to 50 in each class (at intervals of 10). The results of all the models are depicted in Fig. 17, showing a reasonably upward trend with increasing sample sizes. Notably, the rate of performance increase for most models on both datasets is gradually decreasing. For training sets with high intraclass similarity, increasing the number of samples merely enhances repetitiveness. As a result, the model struggles to leverage these repetitive training samples to enhance its comprehensive understanding of ground

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TABLE VIII

QUANTITATIVE COMPARISON OF DIFFERENT STATE-OF-THE-ART CLASSIFICATION ALGORITHMS ON HL DATASET. THE BOLD ENTITIES INDICATE THE BEST ONE

Class.	CNN-based Network				Transformer-based Network							
	SSRN	HybridSN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	morphFormer	CMTFNet	RSSFormer	Proposed
1	36.11	39.93	16.71	37.92	23.96	**52.43**	45.83	26.04	40.62	3.47	36.11	17.36
2	7.46	16.42	80.6	19.4	0	**100**	0	**0**	41.79	100	53.73	85.07
3	71.77	26.38	13.65	31.4	4.61	16.97	34.5	33.58	**74.54**	5.9	11.99	18.63
4	12.66	29.11	25.32	47.09	0	7.59	24.05	22.78	**45.57**	0	12.66	25.32
5	2.28	11.42	0	4.64	2.28	0.21	10.42	0	0	5.14	7.71	**10.49**
6	27.8	18.8	14.2	23.8	26	12.2	11.2	15.4	**39.6**	9.6	0.2	0.6
7	69.92	86.84	59.67	61.98	12.34	**74.69**	15.9	71.84	79.2	11.65	15.92	64.96
8	22.51	12.88	16.8	34.39	50.52	34.07	28.97	0.32	22.15	40.06	58.54	**67.61**
9	**79.95**	67.33	68.27	66.5	55.69	78.96	18.81	40.59	13.12	48.51	10.89	44.8
10	49.9	35.93	63.24	65.91	1.64	4.52	1.23	84.8	5.95	3.9	0.21	**67.15**
11	**100**	90.67	98.78	98.43	99.86	98.78	99.21	**100**	**100**	**100**	88.44	**100**
12	**100**	98.45	**100**	**100**	0	99.78	59.2	**100**	99.33	3.33	46.56	**100**
OA (%)	52.19	52.02	51.56	54.89	30.83	51.89	30.19	45.11	51.85	27.98	33.22	**58.06**
AA (%)	48.36	44.51	46.43	49.29	23.07	48.35	29.11	41.28	46.82	27.63	28.58	**50.16**
Kappa	43.76	42.15	41.89	46.56	17.45	41.7	18.54	36.81	41.56	13.51	19.51	**48.08**

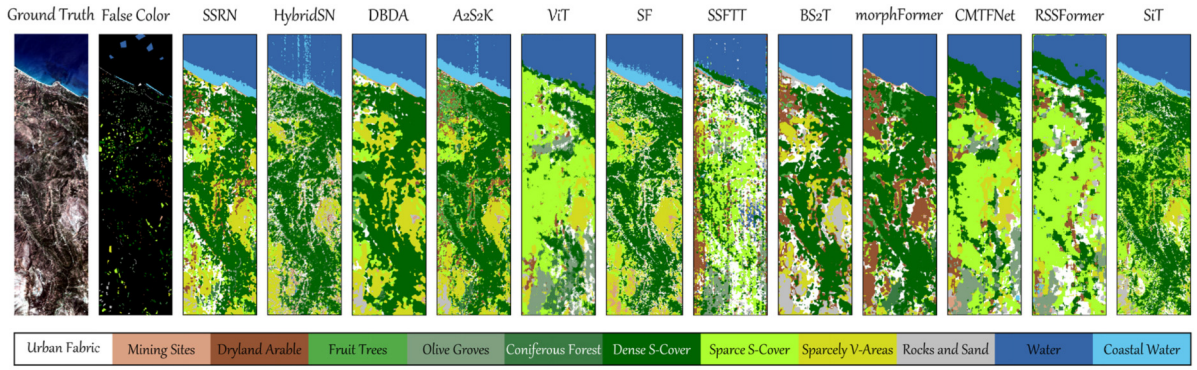


Fig. 19. Visualization of ground-truth map and classification result obtained by different methods on the HL dataset. The best classification map is suggested to be evaluated by jointly comparing the ground-truth image with the false-color image.

cover, making improvements in classification performance difficult to achieve. Moreover, it is worth noting that the total sample size of the PU dataset is much higher than that of the UH dataset, and the proportion of training samples to the total sample size is lower. Nevertheless, our proposed SiT still performs well on this dataset under this low training sample size, despite having no extra advantage in model parameters. Thus, this experimental phenomenon demonstrates that our proposed SiT has significant advantages in feature extraction and model generalization.

H. Evaluation Under Complete Spatial Separation

To better evaluate the generalization of our method, we conduct experiments by training on the HD scene and testing on the HyRANK-Loukia (HL) scene, under a completely spatially separated setting. This setup offers a more rigorous test of robustness and transferability. Details of sample distribution are shown in Table VII and Fig. 18.

Table VIII presents the classification performance of all comparison methods when trained on the HD dataset and tested on the HL dataset. Overall, compared to training and testing on a single dataset, cross-regional evaluation generally results in performance degradation. This performance degradation caused by domain bias is evident when comparing the classification results of water and inland vegetation. All methods exhibit strong domain adaptation for the "Water" class, attributed to the homogeneous spatial structure and low spectral variability of deep-sea areas. Moreover, it is worth noting that, except for our SiT, most transformer-based methods perform worse than CNN-based approaches in this cross-domain setting, particularly those emphasizing spatial feature extraction. Spatial features are more sensitive to scene variations than spectral features, making spatially dependent methods less robust under domain shifts. Furthermore, the weak inductive bias and data-driven nature of transformer architectures tend to overfit local saliency or style features in the training set, which compromises their generalization ability. In contrast, our method incorporates multiscale inductive priors at the shallow level and enforces consistent learning between spatial and spectral features via the proposed S^2C^2 loss, thereby achieving better domain adaptability.

We further provide the classification maps from the above experiments for qualitative analysis (Fig. 19). Most methods perform well in deep-sea areas, consistent with the quantitative results. However, for land surface classification, spatial-feature-dependent methods often suffer from oversmoothing due to significant spatial variations between scenes. In contrast, the spectral characteristics of land covers are more stable across locations. Methods like SF and our SiT, which focus on spectral features, produce more accurate and detailed

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land-cover maps. Although SSRN, HybridSN, and A2S2K also use spectral cues, their hierarchical structures may weaken spectral fidelity during spatial processing.

V. Conclusion

In this article, we introduce SiT, the development of MHSA from the perspective of multiscale feature representation for HSIC tasks. The key idea of this method is to consider heterogeneous scale aspects inside the proposed distinctive SIMHA, in order to precisely identify apparent and discriminative regions of land covers. Moreover, to obtain representative multiscale features for input to the SIMHA, we developed specific feature extraction modules for spatial and spectral features, respectively. Furthermore, a customized loss function is proposed, called S^2C^2 loss function to guide the consistent identification of abstract spatial, spectral, and spatial–spectral features. An extensive experimental analysis implemented on four widely used benchmarks demonstrates the advantages of SiT over the other compared methods for the HSIC task.

In future studies, we will investigate lightweight transformer structures for the requirement of low computational complexity while preserving the accuracy of SiT.

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